

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

Roll No. :S040	Name:ADITYA SOPTE
Class :SY BSC IT	Batch : B2
Date of Assignment :22-07-2026	Date/Time of Submission :22-07-2026

Questions:

2a. Write a python code to create nested tuples.

CODE:

```
GTD.py - C:/Users/IT-23/AppData/Local/Programs/Python/Python314/idle/GTD.py (3.14.6)
File Edit Format Run Options Window Help
student = ("Aditya", 40, ("BSc IT", "Semester III"), 90)

print("Name =", student[0])
print("Roll No =", student[1])
print("Course =", student[2][0])
print("Semester =", student[2][1])
print("Percentage =", student[3])
```

OUTPUT:

```
=====
Name = Aditya
Roll No = 40
Course = BSc IT
Semester = Semester III
Percentage = 90
```

2b. Write a python code to sort the nested tuple using sorted () function.

CODE:

```
*GTD.py - C:/Users/IT-23/AppData/Local/Programs/Python/Python314/idle/GTD.py (3.14.6)*
File Edit Format Run Options Window Help
marks = ((4, 90), (2, 75), (5, 85), (1, 95), (3, 80))

print("Original Tuple:")
print(marks)

Sorted_marks = tuple(sorted(marks))

print("Sorted Tuple:")
print(Sorted_marks)
print("Aditya Sopte S040")|
```

OUTPUT:

```
/Python314/idle/GTD.py =====
Original Tuple:
((4, 90), (2, 75), (5, 85), (1, 95), (3, 80))
Sorted Tuple:
((1, 95), (2, 75), (3, 80), (4, 90), (5, 85))
|
```

2c. Write a python code to copy or clone list.

CODE:

```
*GTD.py - C:/Users/IT-23/AppData/Local/Programs/Python/Python314/idle/GTD.py (3.14.6)*
File Edit Format Run Options Window Help
Num = [10, 20, 30, 40, 50]

lNum = Num.copy()

print("Original List =", Num)
print("Copied List =", lNum)
print("Aditya Sopte S040")|
```

OUTPUT:

```
Original List = [10, 20, 30, 40, 50]
Copied List = [10, 20, 30, 40, 50]
Aditya Sopte S040
```

2d. Write a python code to check immutability property of python tuples.

CODE:

```
GTD.py - C:/Users/IT-23/AppData/Local/Programs/Python/Python314/idle/GTD.py (3.14.6)
File Edit Format Run Options Window Help
d=(40,7)
d[1]="i"
print("Aditya Sopte S040")
```

OUTPUT:

```
===== RESTART: C:/Users/IT-23/AppData/Local/Programs/Python/Python314/idle/GTD.py =====
Traceback (most recent call last):
  File "C:/Users/IT-23/AppData/Local/Programs/Python/Python314/idle/GTD.py", line 2, in <module>
    d[1]="i"
TypeError: 'tuple' object does not support item assignment
>>>
```

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types
- d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
```

```
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks
- Display marks above 80